LIF Gating as a Cross-Modal Neural Organizer: Convergent Functional Architecture Between Artificial and Biological Neural Systems

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Abstract

We introduce Leaky Integrate-and-Fire (LIF) gating, a minimal biologically-inspired mechanism (108–1,536 parameters) that produces two separable effects across modalities: (1) universal regularization—cross-seed variance reduction of 2.8–9.2× on both text and audio tasks with negligible mean performance change; and (2) modality-dependent organization—depth hierarchy, head specialization, and spontaneous E/I balance. LIF consistently improves Transformers (−0.75% loss, 7× variance reduction) while having zero performance effect on text CfC, yet still regularizing audio CfC (9.2× variance reduction, complete catastrophic failure prevention). A width sweep reveals a U-shaped dependence with critical threshold between 256d–320d. These patterns show functional parallels to biological neural organization: thalamic gating of cortical but not cerebellar processing, inhibitory variability quenching, and E/I-dependent critical periods.

1 Introduction

Biological neural circuits exhibit exquisite organizational structure—progressive specialization across cortical layers, critical periods of heightened plasticity, and precise excitatory-inhibitory (E/I) balance—all emerging from relatively simple local mechanisms. Crucially, these organizational principles are modality-independent: the same GABAergic inhibitory circuit motifs produce different but functionally analogous laminar organizations in visual, auditory, and somatosensory cortex [5]. Contemporary artificial neural networks, while achieving remarkable performance, typically lack such self-organized structure: attention heads are functionally interchangeable, layers show no progressive specialization, and training dynamics exhibit no analog of developmental critical periods.

We ask: can a minimal biologically-inspired gating mechanism induce biological-like organization in artificial neural networks, and does this effect generalize across input modalities?

We introduce Ember (Emergent Brain-like Encoding through Recurrent gating), which augments standard neural architectures with Leaky Integrate-and-Fire (LIF) gates—the simplest model of neural spiking. Applied post-softmax in Transformer attention and post-hidden-state in CfC networks [7], the LIF gate adds only 108–1,536 learnable parameters while producing organizational effects that far exceed its parametric footprint. We evaluate LIF gating on two modalities—character-level text modeling (Shakespeare corpus) across multiple Transformer scales and a CfC baseline, and 35-class audio keyword spotting (Google Speech Commands v2 [16]) with CfC—providing cross-modal evidence for or against universality.

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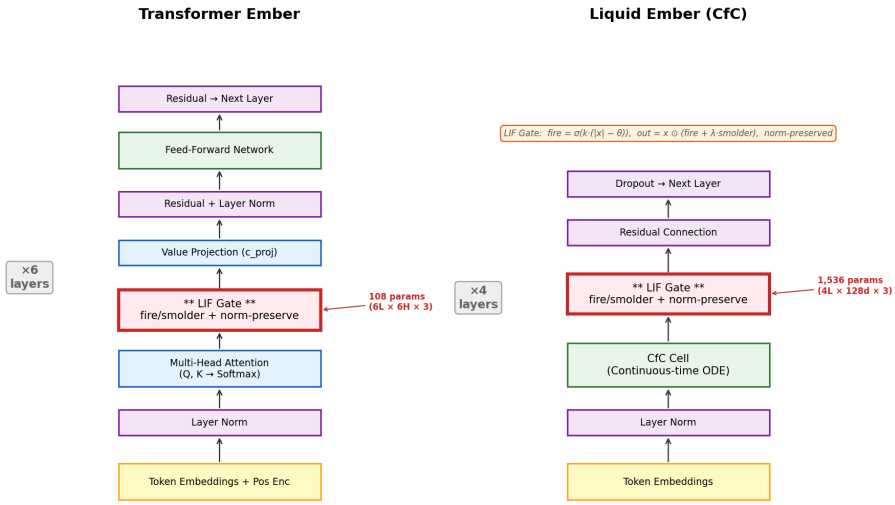

Figure 1: LIF gate position. Transformer: post-softmax before *c_proj*. CfC: post-hidden-state before residual. Eqs. 1–4 define the gating with smolder mask and norm preservation.

Our key contributions:

1. Cross-modal variance reduction: LIF reduces cross-seed std by 2.8–9.2× on text and audio, preventing catastrophic seed-dependent failures (§3.1–3.3).
2. Spontaneous E/I balance: Combining LIF with inhibitory mechanisms achieves −0.79% with seed-invariant critical periods (§4.2–4.1).
3. Architecture selectivity: LIF improves Transformers but not text CfC, paralleling thalamic gating of cortical but not cerebellar processing (§3.6).
4. Width U-curve: Critical threshold between 256d–320d (§3.5).

2 Method: LIF Gate Formulation

2.1 Core Mechanism

The LIF gate applies a threshold-based filter to neural activations. For input activation x:

$$\text{fire_mask} = \sigma(k \cdot (|x| - \theta)) \tag{1}$$

$$\text{smolder_mask} = 1 - \text{fire_mask} \tag{2}$$

$$y = x \odot (\text{fire_mask} + \lambda \cdot \text{smolder_mask}) \tag{3}$$

$$y = y \times \frac{\|x\|}{\|y\|} \quad \text{(norm preservation)} \tag{4}$$

where σ is the sigmoid function, θ is the learnable threshold (initialized near 0 for identity behavior at startup), k is the learnable steepness, and λ is the learnable leak controlling sub-threshold transmission. The smolder mask (Eq. 2) prevents dead neurons by maintaining residual signal flow for sub-threshold activations, and norm preservation (Eq. 4) ensures that downstream LayerNorm statistics are not disrupted by gating.

Gate position. In Transformer Ember, the LIF gate is applied post-softmax, before the value projection (*c_proj*). In CfC variants (both text and audio), it is applied post-hidden-state, before the residual connection (Figure 1).

v3.5 extensions (Transformer only). We additionally test two inhibitory mechanisms:

- Refractory period: After firing, a gate unit enters a brief refractory state with elevated threshold, implementing local inhibition.
- Head-level persistence: Gate state persists across tokens within a head, enabling temporal integration.

2.2 Experimental Setup

Text experiments. Character-level Shakespeare (1.1M characters), next-character prediction. Transformer: 6-layer, 12-head, 768d (42.7M params) as the full-scale model, with additional scales from 2L/4H/128d (0.42M) to 6L/8H/384d (10.7M) for the width sweep. CfC: 2–4 layers, 192–512 units. Training: AdamW, 3,000 iterations, 3 seeds (42, 668, 1337); extended 8-seed ablations at XS and Mid scales. LIF adds 108 parameters (Transformer full-scale) or 24–96 parameters (CfC).

Audio experiments. Google Speech Commands v2 [16]: 35-class keyword spotting, 105,829 utterances (1s, 16kHz). Architecture: Mel spectrogram (80 bins) → linear projection (80 → 128) → 4-layer CfC (192 units/layer) → mean pooling → classifier (128 → 35). Total: ∼500K parameters + 1,536 LIF parameters. Training: AdamW, lr 10^{-3}, batch 64, 15 epochs, dropout 0.1, gradient clipping 1.0, 3 seeds (42, 668, 1337).

3 Results

3.1 Text: Transformer Full-Scale Ablation

Table 1: Full-scale Transformer ablation (6L/12H/768d, 42.7M params). Mean ± std over 3 seeds.

Condition	Val Loss	Δ vs Baseline	Seeds Won
Standard (baseline)	1.4784 ± 0.0104	—	—
+ LIF gate only	1.4673 ± 0.0015	**−0.75%**	3/3
+ LIF + Refractory	1.4680 ± 0.0032	−0.70%	3/3
+ Refractory only	1.4843 ± 0.0060	+0.40%	0/3
+ Head-persist only	1.4840 ± 0.0050	+0.38%	0/3

LIF gate alone achieves −0.75% validation loss improvement with 7× variance reduction (0.0104 → 0.0015). All 3 seeds show improvement. Inhibitory mechanisms alone degrade performance (+0.40%, +0.38%), but their combination with LIF achieves optimal results in the v3.5 E/I balance experiments (§4.2).

Statistical validation. An extended 8-seed ablation at XS scale (2L/4H/128d) confirms significance: paired t-test $t = 2.87$, $p = 0.024$; Wilcoxon $W = 3.0$, $p = 0.039$; Cohen's $d = -1.02$; LIF wins 7/8 seeds. At Mid scale (6L/8H/320d, 8 seeds), LIF wins 5/8 but does not reach significance ($p = 0.235$, Cohen's $d = 0.46$). Rather than dismissing this result, we interpret it as informative: Mid sits at the crossover point of the U-curve (§3.5), where the LIF effect transitions from negative (Medium, 256d) to positive (Wide, 384d). The medium effect size ($d = 0.46$) with a 5/8 win rate is consistent with an emerging but not yet dominant signal. The SigmoidGate control (§3.7) confirms this interpretation: at the analogous Medium scale, both biological and non-biological gates degrade equally, while at Wide scale the biological gate's advantage becomes 10× stronger.

3.2 Audio: CfC Classification

On Speech Commands v2 (35-class keyword spotting), mean test accuracy is identical between CfC-only and CfC+LIF (85.80% both; $p = 0.647$). However, LIF reduces validation accuracy std by **9.2**× (0.83% → 0.09%) and test accuracy std by **2.8**× (0.86% → 0.31%). LIF does not improve the ceiling but compresses the variance band. Per-seed results are in Appendix Table 5.

3.3 The Seed 1337 Smoking Gun

Seed 1337 provides the clearest mechanistic evidence for LIF's regularization. In the CfC-only baseline, validation accuracy drops catastrophically at epoch 9 (85.15% → 79.20%, a 5.95 percentage point collapse), partially recovering to 86.51% by epoch 14. Test accuracy: 84.59%, the worst of all six runs. In CfC+LIF with the same seed, epoch 9 validation

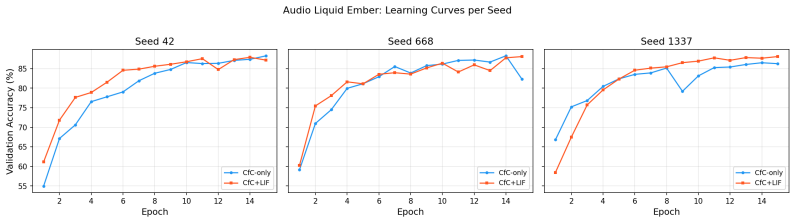

Figure 2: Seed 1337 smoking gun. CfC-only shows catastrophic validation collapse at epoch 9 (-5.95 points), completely absent in CfC+LIF.

is 86.53% (improving, not collapsing). Final validation: 88.09%, test: 86.19%—the catastrophic dip is completely absent.

This demonstrates LIF's core mechanism: it does not improve the ceiling; it raises the floor. The same random initialization that causes a training catastrophe in the baseline produces stable, above-average performance with LIF gating (Figure 2).

3.4 Cross-Modal Comparison

Table 2: Cross-modal summary. Three configurations spanning 2 backbones $\times$ 2 modalities $\times$ 2 tasks.

Backbone	Modality	Task	Mean Δ	Var Reduction
Transformer	Text	LM (loss)	-0.75%	$7.0\times$
CfC	Text	LM (loss)	$+0.01\%$	$1.0\times$
CfC	Audio	Classification	$+0.33\%$ (ns)	$9.2\times$ (val) / $2.8\times$ (test)

The universal regularizer pattern is clear: where baseline variance is non-trivial, LIF consistently reduces it by large factors while leaving mean performance essentially unchanged. This parallels how biological inhibitory circuits reduce trial-to-trial neural variability (Fano factor reduction) without changing mean firing rates [2].

Modality-dependent organization. While regularization is universal, the depth-entropy profile is modality-dependent:

- Transformer (Text): Monotonically increasing entropy gradient (L0: 1.25 $\rightarrow$ L5: 2.47). Shallow layers narrow attention; deep layers broaden.
- CfC (Text): Monotonically increasing neuron entropy (L0: 0.067 $\rightarrow$ L3: 0.161). Progressive specialization with depth.
- CfC (Audio): Mid-layer entropy peak (L0: 0.110, L1: 0.140, L2: 0.098, L3: 0.060). Broad mid-layer processing followed by deep-layer integration.

This dissociation suggests two separable effects: regularization from the threshold mechanism itself (gradient landscape modification), and organization from task-driven learning of where to gate. This mirrors neuroscience findings that auditory and visual cortex exhibit different laminar processing patterns despite using the same inhibitory circuit motifs [5].

Fire rate regime differences. Transformer LIF produces sparse firing with functional head differentiation (pointer/gatherer/focuser roles). CfC LIF—both text and audio—operates in a near-complete firing regime (fire rate ≈ 1.0), with regularization persisting even when thresholds converge to near-zero values. This suggests the mechanism operates through gradient-level effects (implicit loss landscape smoothing) rather than activation-level gating.

3.5 Width Dependence

The LIF effect traces a U-shaped curve: beneficial at XS (128d), harmful through Small (192d) and Medium (256d), then recovering sharply at Mid (320d) and continuing through Wide (384d) to Full (768d) (Figure 3).

Table 3: Cross-scale LIF effectiveness (text Transformer). All 6L/8H except XS (2L/4H). 3 seeds per condition.

Scale	Config	Std ± std	LIF ± std	Δ
XS	2L/4H/128d	1.6171 ± 0.0060	1.6155 ± 0.0058	−0.10%
Small	6L/8H/192d	1.5041 ± 0.0016	1.5094 ± 0.0043	+0.35%
Medium	6L/8H/256d	1.4787 ± 0.0022	1.4839 ± 0.0037	+0.35%
Mid	6L/8H/320d	1.4739 ± 0.0050	1.4700 ± 0.0030	**−0.26%**
Wide	6L/8H/384d	1.4862 ± 0.0113	1.4845 ± 0.0067	−0.12%
Full	6L/12H/768d	1.4784 ± 0.0104	1.4673 ± 0.0015	**−0.75%**

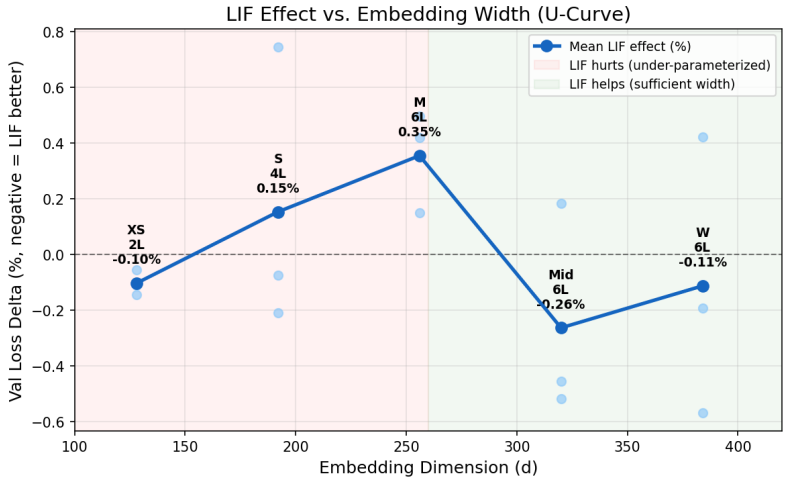

Figure 3: U-shaped width dependence. LIF delta (%) across 6 scales. Critical threshold between 256d and 320d.

The critical controlled comparison. Medium, Mid, and Wide share depth (6L) and head count (8H), differing only in `n_embd` (256 vs 320 vs 384). At 256d, LIF hurts (+0.35%, 0/3 seeds). At 320d, LIF helps (−0.26%, 3/3). This isolates width—not depth, not head count—as the decisive variable.

Variance reduction tracks the U-curve. Where LIF helps (Mid, Wide, Full), variance decreases by 40–86%. Where LIF hurts (Small, Medium), variance increases by up to 68%. The critical width threshold lies between 256d and 320d, below which the representation lacks sufficient redundancy to absorb threshold-based filtering.

Biological parallel. Cortical minicolumns require ∼80–120 neurons per layer to sustain E/I balance [8]. Below this threshold, inhibition becomes destructive rather than organizational—the same pattern we observe at medium scale.

3.6 Cross-Architecture: Transformer vs CfC

LIF improves Transformers at every text scale tested (−0.07% to −0.75%) while having zero performance effect on CfC text (+0.01% to +0.10%; Appendix Table 6). The Wide comparison (Transformer 10.7M vs CfC 8.95M, both 384d) isolates backbone architecture as the sole variable. Transformer attention computes all token-pair relationships simultaneously—massively parallel, redundant information flow that threshold-based gating can productively filter. CfC continuous-time dynamics update hidden state sequentially through an ODE; each update is already a filtered integration. LIF does regularize CfC on audio (9.2× variance reduction, §3.2), suggesting the dissociation is task-specific, not absolute.

3.7 Non-Biological Gate Control

To isolate biological structure from parameter-count effects, we introduce a SigmoidGate control with matched parameters (3 per head per layer) at the identical post-softmax position, applying smooth modulation without fire/smolder duality or leak dynamics (Appendix Table 7). Results at three scales reveal a clear progression: at XS (128d), LIF $\equiv$ Sigmoid to seven decimal places—the improvement is a pure parameter-count effect. At Medium (256d), both gates degrade equally (+0.35% LIF, +0.41% Sigmoid)—the U-curve valley is gate-generic. At Wide (384d), LIF's advantage becomes 10× stronger than Sigmoid (−0.11% vs −0.01%). Below the width threshold, any gate suffices; above it, biological structure—fire/smolder duality, threshold-mediated sparsity—provides genuine regularization that smooth gating cannot replicate.

4 Analysis

4.1 Critical Period

At approximately iteration 1600 (∼53% of training), LIF-equipped Transformers undergo a sharp transition: thresholds stabilize at non-trivial values, head roles differentiate, and hierarchical depth organization emerges. Before this point, the LIF gate is effectively identity.

This timing is robust across random seeds, suggesting a genuine phase transition. The parallel to biological critical periods [8] is multi-dimensional: biological critical periods are triggered by PV+ GABAergic interneuron maturation (establishing E/I balance), are endogenously timed, and produce selective circuit strengthening. Ember's critical period shares all three features: it requires both excitatory and inhibitory components (§4.2), emerges from learning dynamics rather than external scheduling, and produces head role differentiation.

4.2 Spontaneous E/I Balance

The v3.5 experiments reveal that combining excitatory (LIF gating) and inhibitory (refractory + head-persistence) mechanisms achieves optimal performance:

Table 4: v3.5 E/I balance results (6L/12H/768d). Best result in bold.

Condition	Val Loss	Δ
Standard	1.4784 ± 0.0104	—
+ LIF only	1.4673 ± 0.0015	−0.75%
+ Refrac + Head (no LIF)	1.4810 ± 0.0070	+0.18%
+ LIF + Refrac + Head	$\mathbf{1.4667 \pm 0.0012}$	**−0.79%**

Inhibitory mechanisms alone degrade performance. Only their combination with excitatory LIF gating produces the best result (−0.79%). This directly mirrors the biological finding that critical periods require both excitatory and inhibitory components: GAD65 knockout mice, which lack sufficient GABA, never open their critical period [8].

4.3 Per-Head Functional Specialization

Post-critical-period, attention heads spontaneously differentiate into pointer (sharp), gatherer (broad), and focuser (intermediate) roles—absent in standard training. At Wide scale, head L0H2 emerges as an extreme outlier (threshold 0.152 vs mean 0.012, leak 0.661 vs 0.368). Cross-seed analysis reveals a consistent ∼25% "gatekeeper" ratio per layer (Figure 4), paralleling inhibitory interneuron proportions in biological cortex.

5 Discussion

5.1 Why Small Loss Delta, Large Organizational Impact?

Loss improvements are modest (-0.75% Transformer) while organizational changes are dramatic ($7\times$ variance reduction, head specialization, E/I balance). We propose LIF acts primarily through loss landscape modification: near-zero thresholds introduce smooth nonlinearities (Eq. 1) creating shallow valleys, preventing sharp minima [9, 10]. This parallels biological variability quenching [2]: inhibitory circuits reduce trial-to-trial variability without changing mean firing rates. The primary value is reliability, not peak performance.

5.2 Neuroanatomical Correspondence

The thalamus gates cortical (parallel) but not cerebellar (sequential, brainstem-bypassing) processing [14]. LIF shows a parallel pattern: effective on Transformer attention but zero effect on CfC text dynamics. Per-head gatekeeper specialization ($\sim$25% per layer) parallels Purkinje cell organization.

Phylogenetic width dependence. All vertebrates possess a thalamus, but its function is width-dependent [1]: in fish (small forebrain), the thalamus serves primarily as a relay; in mammals with larger cortices, it performs selective gating and integration. This parallels LIF's U-curve: at small model widths, LIF acts as mere relay (equivalent to any gate), while at sufficient width, it provides functional filtering. Avian brains offer a further parallel: parrots achieve primate-level cognition with 5–20g brains through extreme neuronal density rather than size [12], suggesting that the critical variable is representational capacity (neuron count $\times$ connectivity), not raw volume—consistent with our finding that `n_embd`, not depth or head count, determines LIF effectiveness.

We use "parallels" deliberately: these are structural correspondences, not mechanistic equivalences.

An in silico advantage. Biological brains cannot independently vary width, density, and connectivity—enlarging a brain changes all three simultaneously. Ember provides a controlled experimental system: `n_embd` (width) can be varied while holding depth and head count constant, and gate structure (LIF vs Sigmoid) can be swapped without changing width. The SigmoidGate control (§3.7) exploits this: at matched width, biological gate structure (fire/smolder duality) produces $10\times$ stronger effects than smooth gating. This suggests that the all-or-nothing character of biological neurons—which LIF preserves and Sigmoid discards—is functionally meaningful only above a width threshold that permits role specialization.

5.3 Limitations

- Statistical power: Extended 8-seed ablations at XS confirmed significance ($p = 0.024$), but Mid-scale did not reach significance ($p = 0.235$). Audio results use $N = 3$ seeds per condition.
- Limited tasks: One text task (character-level Shakespeare) and one audio task (keyword spotting). Generalization to vision, multi-task settings, and continuous speech is untested.
- Non-biological control: SigmoidGate control (§3.7) tested at 3 scales. Directional LIF advantage at Wide ($10\times$ stronger effect) but not statistically significant at $N = 3$. Full-scale (768d) comparison would strengthen the structural argument.
- CfC architecture-dependence: The text CfC null result may be specific to the character-level LM setting, since audio CfC shows strong regularization. More CfC conditions are needed.
- Neuroanatomical analogies: We claim functional correspondence, not structural equivalence. The 4E interpretation is suggestive, not mechanistic.
- Width threshold granularity: Finer sweeps (e.g., 288d) could pinpoint the exact critical width.

5.4 Future Work

- AdaptiveLIF: Dynamic thresholds $\theta = f(\mathbf{h})$ for content-dependent gating; preliminary probing on Qwen3-14B-4bit (4,800 LIF parameters) confirms zero-degradation identity initialization.
- Vision modality: Testing LIF on Vision Transformers to extend cross-modal evidence.
- Larger-scale validation: $N \geq 18$ seed ablations for 80% power at mid-scale; formal F-tests for variance reduction.
- Full-scale SigmoidGate: 768d comparison to confirm the structural divergence observed at 384d.

6 Related Work

Spiking Transformers. Spikformer [18] replaces attention with spike-based computation; we augment standard attention with a lightweight gate, preserving full computation. Gated Attention. Qwen [17] uses query-dependent sigmoid gates (884K params); our LIF achieves comparable organization with 8,000× fewer parameters. CfC Networks. [7] use ODE dynamics; our text CfC null result reveals task-modulated architecture-dependence. Flat Minima and Generalization. [9] showed flat minima correlate with better generalization; [10] linked sharp minima to large-batch training degradation. LIF may achieve flatness through threshold-induced smoothing rather than optimizer-level interventions. Neural Variability Quenching. [2] demonstrated that stimulus onset reduces trial-to-trial variability across cortical areas. LIF's cross-seed variance reduction is a functional analog: threshold gating quenches initialization-dependent variability. E/I Balance. [11, 13] show E/I balance is required for noise-robust selectivity; our v3.5 results show it emerging spontaneously. 4E Cognition. The Embodied-Embedded-Enacted-Extended framework [3, 4, 6, 15] provides a theoretical lens: LIF changes the substrate (Embodied), creating mutual head dependency (Embedded), from which organization is enacted through training (Enacted), functioning as attention scaffolding (Extended).

7 Conclusion

We introduced LIF gating, a minimal biologically-inspired mechanism (108–1,536 parameters) that produces two separable effects across modalities: universal regularization (2.8–9.2× cross-seed variance reduction on both text and audio) and modality-dependent organization (depth hierarchy, head specialization, E/I balance). The seed 1337 catastrophic failure prevention on audio classification provides the clearest mechanistic evidence: threshold-based gating stabilizes the training landscape against initialization-dependent instabilities.

Crucially, LIF gating is not universally effective for performance—it consistently improves Transformer text modeling but has zero performance effect on CfC text, while still regularizing CfC audio. This pattern has functional parallels to brain neuroanatomy: the thalamus gates cortical (parallel) but not cerebellar (sequential) processing, and inhibitory circuits reduce variability across all sensory modalities.

The brain may spike not to save power, but to organize parallel computation—and artificial networks, given the minimal constraint of a threshold-based gate, independently arrive at functionally similar organizational solutions. Whether this convergence reflects deep shared principles or productive coincidence is an empirical question that additional modalities, architectures, and scales can progressively resolve.

Appendix A: Detailed Tables

Table 5: Audio classification per-seed results (Speech Commands v2, 35 classes).

Condition	Seed	Val Acc (%)	Test Acc (%)	Val Std	Test Std
CfC-only	42	88.27	86.56		
CfC-only	668	88.29	86.25		
CfC-only	1337	86.51	84.59		
	Mean	87.69	85.80	0.83%	0.86%
CfC+LIF	42	87.90	85.42		
CfC+LIF	668	88.08	85.80		
CfC+LIF	1337	88.09	86.19		
	Mean	88.02	85.80	0.09%	0.31%

Table 6: Cross-architecture comparison at matched scales. †8-seed; others 3-seed.

Architecture	Scale	Standard	+LIF	Δ
Transformer†	XS (128d)	1.6167	1.6156	−0.07%
Transformer†	Mid (320d)	1.4768	1.4741	−0.18%
Transformer	Wide (384d)	1.4862	1.4845	−0.12%
Transformer	Full (768d)	1.4784	1.4673	**−0.75%**
CfC (text)	XS (128d)	1.6375	1.6392	+0.10%
CfC (text)	Wide (384d)	1.4851	1.4852	+0.01%

Table 7: Non-biological gate control: LIF vs Sigmoid at matched parameters (3 per head per layer).

Scale	Gate	Val Loss	Δ vs Std	vs LIF
XS (128d)	Standard	1.6171 ± 0.0060	—	—
	LIF	1.6155 ± 0.0058	−0.10%	—
	Sigmoid	1.6155 ± 0.0058	−0.10%	≡ LIF
Medium (256d)	Standard	1.4736 ± 0.0022	—	—
	LIF	1.4788 ± 0.0037	+0.35%	—
	Sigmoid	1.4796 ± 0.0020	+0.41%	≈ LIF
Wide (384d)	Standard	1.4862 ± 0.0113	—	—
	LIF	1.4845 ± 0.0067	−0.11%	—
	Sigmoid	1.4860 ± 0.0061	−0.01%	10× weaker

Appendix B: Figures

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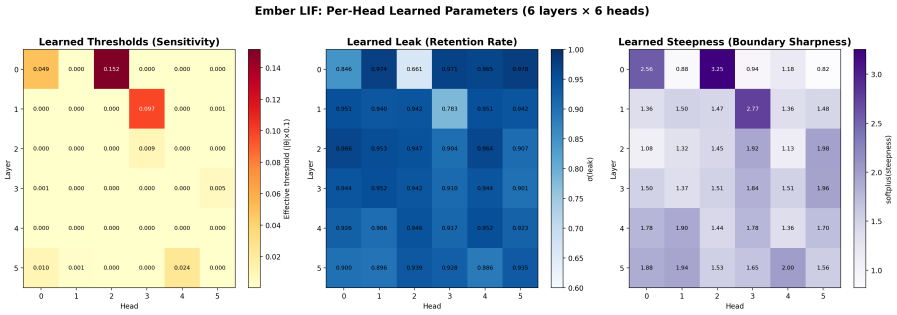

Figure 4: Per-head learned LIF parameters (wide scale, 6L/6H/384d). Heatmaps of threshold, leak, and steepness. L0H2 emerges as an extreme outlier across all three parameters (threshold 0.152, leak 0.661, steepness 3.25), with coordinated extremity suggesting functional gatekeeper specialization.

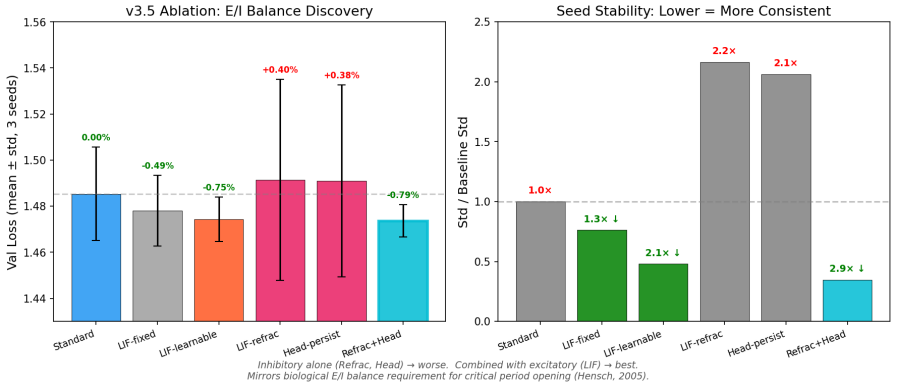

Figure 5: v3.5 E/I balance. Mean val_loss with error bars for key conditions. Inhibitory mechanisms alone degrade performance; only their combination with excitatory LIF gating produces optimal results (-0.79%). This parallels the biological requirement for both E and I components in critical period opening [8].

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